

Modules and Packages in Python

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Modules in Python

Modules: Introduction

A **module** in Python is a file containing Python function and class definitions. The filename has **.py** extension.

💡 Modules are used to organize related code into a single file that can be reused across multiple programs.

Creating a Module



1. Create a new .py file (e.g., **my_module.py**)
2. Write your Python code in this file
3. Save the file

Scenario: Without Modules

Redford school is tasked with developing a school management system having these functionalities:
student management, teacher management, fee calculation and grading system.

School management.py
(single file containing the entire code)



```
def add_student(name, age, grade):
    """Add a new student to the system."""
    # ...python code...
    print(f"Added student: {name}, age: {age}, grade: {grade}")

def update_student(student_id, **kwargs):
    """Update student information."""
    # ...python code...
    print(f"Updated student {student_id} with {kwargs}")

def get_student(student_id):
    """Retrieve student information."""
    # ...python code...
    print(f"Retrieved information for student {student_id}")

def add_teacher(name, subject):
    """Add a new teacher to the system."""
    # ...python code...
    print(f"Added teacher: {name}, subject: {subject}")

def assign_class(teacher_id, class_id):
    """Assign a teacher to a class."""
    # ...python code...
    print(f"Assigned teacher {teacher_id} to class {class_id}")

def get_teacher_schedule(teacher_id):
    """Retrieve a teacher's schedule."""
    # ...python code...
    print(f"Retrieved schedule for teacher {teacher_id}")
```

```
calculate_tuition(student_id, semester):
    """Calculate tuition fees for a student."""
    # ...python code...
    print(f"Calculated tuition for student {student_id} for semester {semester}")

apply_scholarship(student_id, scholarship_amount):
    """Apply a scholarship to a student's account."""
    # ...python code...
    print(f"Applied ${scholarship_amount} scholarship to student {student_id}")

generate_invoice(student_id):
    """Generate an invoice for a student."""
    # ...python code...
    print(f"Generated invoice for student {student_id}")

def record_grade(student_id, subject, grade):
    """Record a grade for a student in a specific subject."""
    # ...python code...
    print(f"Recorded grade {grade} for student {student_id} in {subject}")

def calculate_gpa(student_id):
    """Calculate the GPA for a student."""
    # ...python code...
    print(f"Calculated GPA for student {student_id}")

def generate_report_card(student_id):
    """Generate a report card for a student."""
    # ...python code...
    print(f"Generated report card for student {student_id}")
```

Creating Modules

Redford school is tasked with developing a school management system having these functionalities:
student management, teacher management, fee calculation and grading system.

school_management.py

```
def add_student(name, age, grade):
    """Add a new student to the system."""
    # ...python code...
    print(f"Added student: {name}, age: {age}, grade: {grade}")

def update_student(student_id, **kwargs):
    """Update student information."""
    # ...python code...
    print(f"Updated student {student_id} with {kwargs}")

def get_student(student_id):
    """Retrieve student information."""
    # ...python code...
    print(f"Retrieved information for student {student_id}")

def add_teacher(name, subject):
    """Add a new teacher to the system."""
    # ...python code...
    print(f"Added teacher: {name}, subject: {subject}")

def assign_class(teacher_id, class_id):
    """Assign a teacher to a class."""
    # ...python code...
    print(f"Assigned teacher {teacher_id} to class {class_id}")

def get_teacher_schedule(teacher_id):
    """Retrieve a teacher's schedule."""
    # ...python code...
    print(f"Retrieved schedule for teacher {teacher_id}")
```

teacher_management.py

fee_calculation.py

```
calculate_tuition(student_id, semester):
    """Calculate tuition fees for a student."""
    # ...python code...
    print(f"Calculated tuition for student {student_id} for semester {semester}")

apply_scholarship(student_id, scholarship_amount):
    """Apply a scholarship to a student's account."""
    # ...python code...
    print(f"Applied ${scholarship_amount} scholarship to student {student_id}")

generate_invoice(student_id):
    """Generate an invoice for a student."""
    # ...python code...
    print(f"Generated invoice for student {student_id}")

def record_grade(student_id, subject, grade):
    """Record a grade for a student in a specific subject."""
    # ...python code...
    print(f"Recorded grade {grade} for student {student_id} in {subject}")

def calculate_gpa(student_id):
    """Calculate the GPA for a student."""
    # ...python code...
    print(f"Calculated GPA for student {student_id}")

def generate_report_card(student_id):
    """Generate a report card for a student."""
    # ...python code...
    print(f"Generated report card for student {student_id}")
```

grading_system.py

Using Modules

Redford school is tasked with developing a school management system having these functionalities:
student management, teacher management, fee calculation and grading system.



main.py

```
##### File: main.py #####
# Python Script where the School Management modules are being used

# First Import all functions from these 4 modules
import student_management
import teacher_management
import fee_calculation
import grading_system

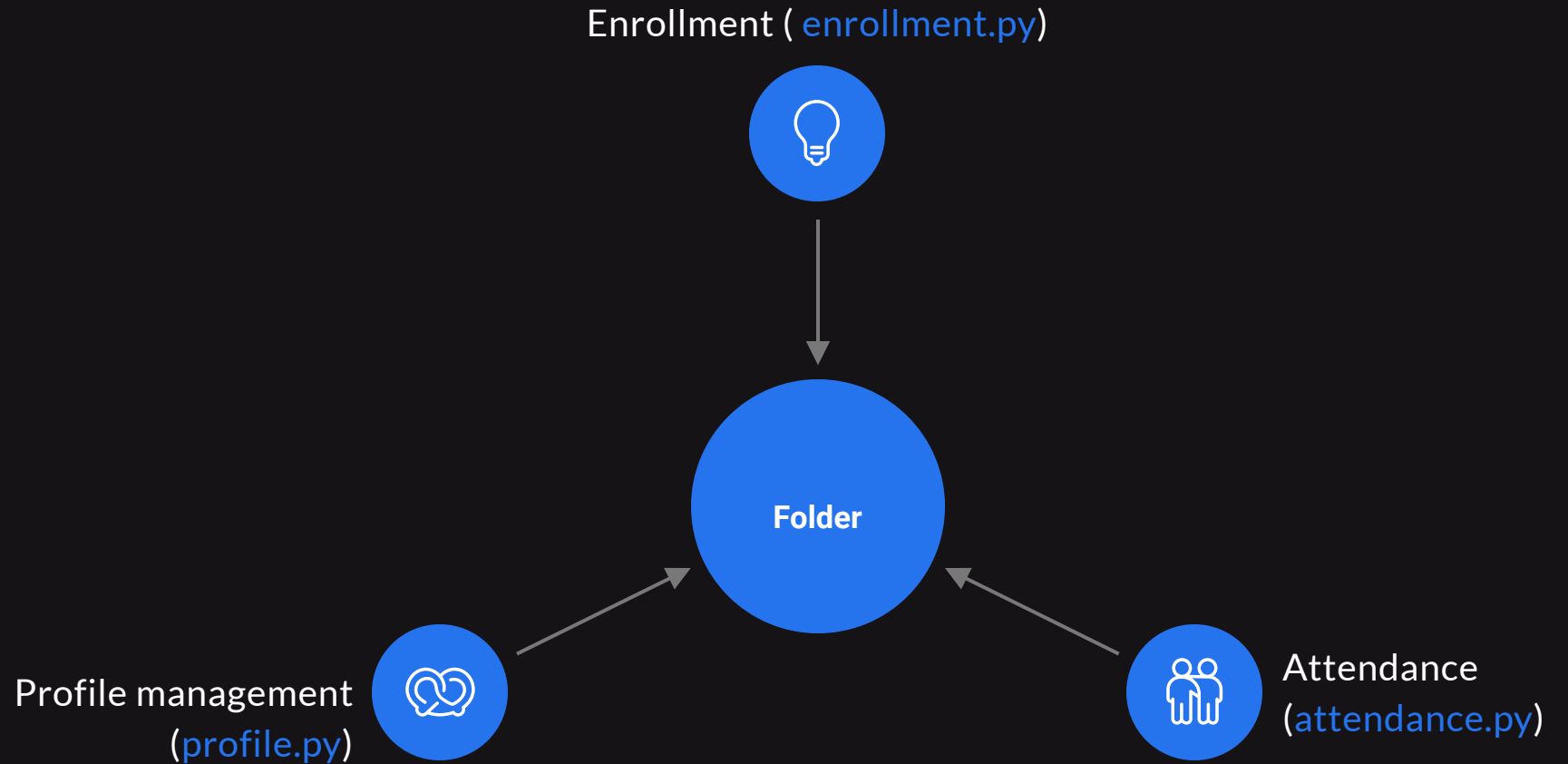
# Example usage of the school management system
student_management.add_student("Alice Smith", 15, 10)
teacher_management.add_teacher("Mr. Johnson", "Mathematics")
fee_calculation.calculate_tuition(1001, "Fall 2024")
grading_system.record_grade(1001, "Mathematics", "A")
```

School notices as Management System grows, breaking down the code into modules is no longer sufficient.

Packages in Python

Scenario: Packages

Student Management folder contains these three modules:



Packages: Introduction

In Python, **package** is a folder containing one or more module files; additionally a special file "**`__init__.py`**".

💡 Note: Packages can also contains sub-packages (nested directories with their own `__init__.py` files)

School Management Package
Directory Structure



```
school_management/  
  __init__.py  
  student_management/  
    __init__.py  
    enrollment.py  
    attendance.py  
    profile.py  
  teacher_management/  
    __init__.py  
    assignments.py  
    scheduling.py  
  fee_management/  
    __init__.py  
    tuition.py  
    extracurricular.py
```

Student Management Sub-Package

Teacher Management Sub-Package

Fee Management Sub-Package

Packages: Usage



Package name	Sub-Package name	Module name	Function name
<pre># main.py</pre>			
<code>from school_management</code>	<code>.student_management</code>	<code>.enrollment</code>	<code>import enroll_student</code>
<pre>from school_management.fee_management.tuition import calculate_tuition</pre>			
<pre># Using functions from the package</pre>			
<pre>print(enroll_student("Alice", 14))</pre>			
<pre>print(calculate_tuition("Alice", 5))</pre>			

Using the packages in the main script

Thank You
